

HW0 LA Review in R

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```
library('ggplot2')  
library('dplyr')
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

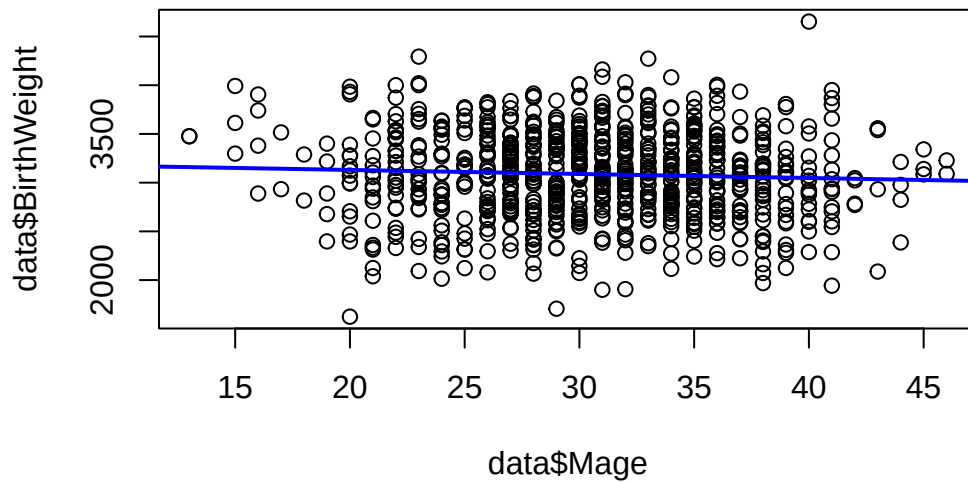
The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
data <- read.table("BirthWeights.txt", sep = " ", header = TRUE, as.is = FALSE)
```

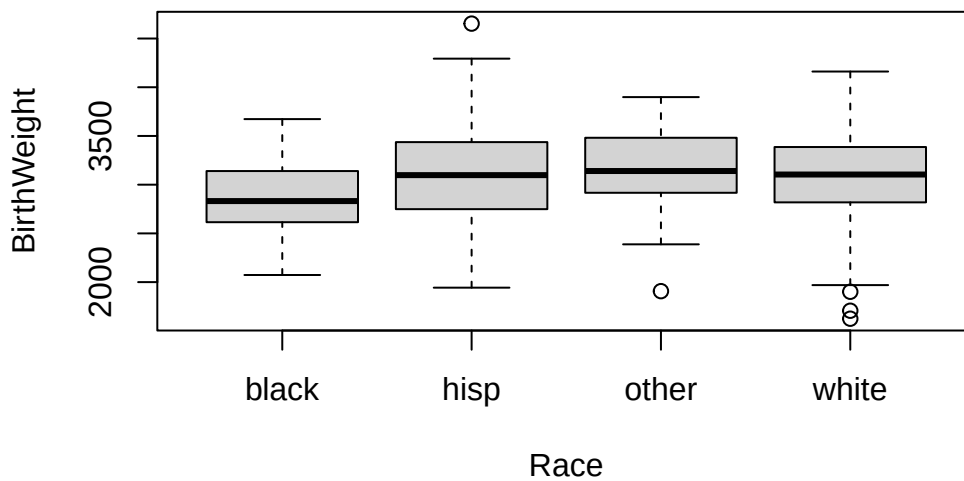
1.

```
lm <- lm(BirthWeight ~ Mage, data = data)  
  
plot(data$Mage, data$BirthWeight)  
abline(lm, col="blue", lwd=2)
```



2.

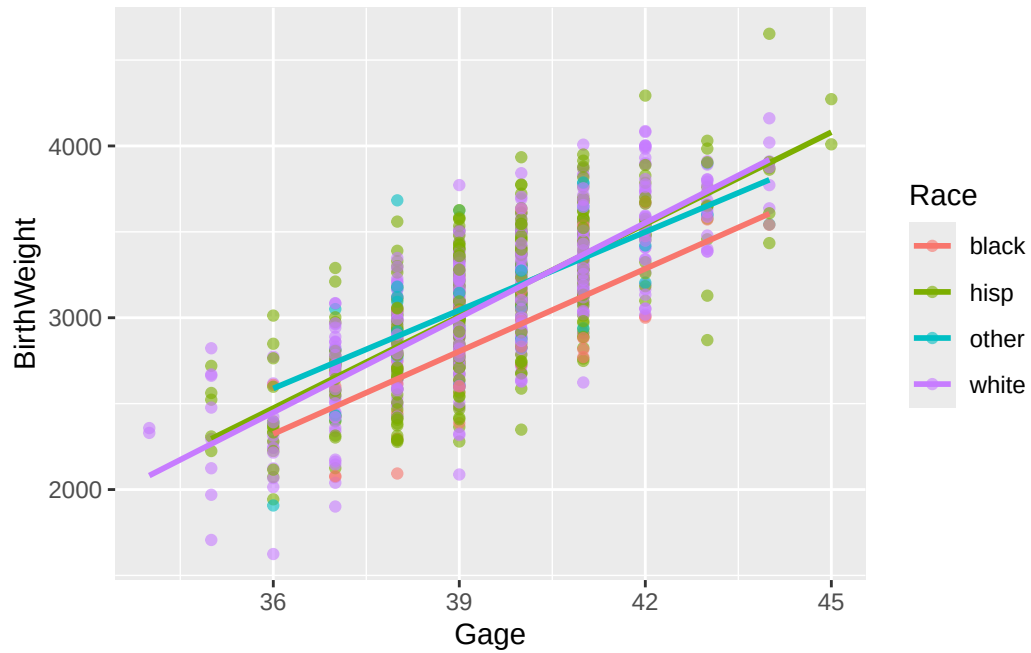
```
boxplot(BirthWeight ~ Race, data = data)
```



3.

```
ggplot(data, aes(x = Gage, y = BirthWeight, color = Race)) +  
  geom_point(alpha = 0.6) + # Plot scatter points with slight transparency  
  geom_smooth(method = "lm", se = FALSE) # Add linear model lines without standard error shading
```

```
`geom_smooth()` using formula = 'y ~ x'
```



4.

```
data %>%  
  group_by(Gen) %>%  
  summarise(correlation = cor(BirthWeight, Mage))
```

```
# A tibble: 2 x 2  
  Gen      correlation  
  <fct>      <dbl>  
1 Female -0.0261  
2 Male   -0.0745
```

5.

```
GGally::ggpairs(data)
```

```

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